ARTEMIS MISSION DATA EXPLORER

Group 2

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PROBLEM STATEMENT

Accomplish: Easy access to the public to find out information about the Artemis missions

Important: Critical for advancing in deep space exploration and a stepping stone for future human missions to Mars

Interest: Both have an interest in space exploration

PROJECT DESCRIPTION

- ★ Designed to make **NASA Artemis** mission **easy** to understand.
- ★ Uses a **menu system** so users can explore:
  - Artemis I, II, III Mission overview
  - Artemis I, II, III Mission Trajectory Visualization
  - Why lunar missions matter?
  - Entire Artemis program summary
  - Mission timeline visualization (1-7 Projected)
- ★ Includes links to **Artemis news** and **NASA image gallery**
- ★ **Goal:** Present *complex* mission data in a simple, organized and user-friendly format for public use.

TECHNICAL CHALLENGES & SOLUTION

- ★ Large amount of mission data was hard to organize
- ★ Needed an easy to used interface
- ★ Hard to visualize mission timeline
- ★ Connecting users to external resources

- ★ Create a structured missionData array to store all mission data
- ★ Implemented menu system
- ★ Used graphs & Plot functions
- ★ Added function to open NASA websites

INCLASS IMPLEMENTATION

Functions: (`buildMissionData`, `showMissionSummary`, etc.)
Structures : (`missionData`)
Loops: Used a `while` loop to keep the program running until the user chooses to exit
Conditional Statements: Used `switch` statements to control what happens based on the user's menu selection
User Interface (Menu): Implemented MATLAB's `menu()` function to create an interactive user experience
Data Visualization: Created plots using `stem()` and `bar()` to visualize the mission timeline and crew sizes
Error Handling: Used `try/catch` to safely handle issues when opening external websites
String Formatting: Used `sprintf()` to neatly display mission information in dialog boxes

DEMO

ARTEMIS